

FINAL EXPLANATION: BIOLOGY GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: BIOLOGY GRADE 10 – 1.3 CELL BIOLOGY: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving Question: Why do the cells in my body, and in the sukuma wiki on my plate, look so different from each other, and how does each cell's structure allow it to do its specific job? Remember the Kisumu blood smear: red blood cells with no nucleus next to white blood cells with large nuclei, and the sukuma wiki leaf cell that is rectangular, has chloroplasts and a large central vacuole. Answer every section in the space under each prompt. Write in full sentences and use the scientific words you learned in Lessons 1 to 8. Use evidence from the images, practicals and activities we did. Where a prompt asks for a Claim, Evidence and Reasoning (CER) answer, state your claim, back it with evidence, and then explain why the evidence supports your claim. Work on your own.

Part 1: What do all the cells share?

The red blood cell, the white blood cell and the sukuma wiki leaf cell all look very different, yet we call all of them cells. Bacteria in fermenting uji are also called cells. (a) Name the structures or features that ALL cells share. (b) Use a Claim, Evidence and Reasoning (CER) paragraph to explain why bacteria count as true cells even though they have no membrane-bound nucleus. (c) Why do you think these shared features are essential for life?

(a) All cells have a cell membrane, cytoplasm, genetic material (DNA) and ribosomes. (b) Claim: Bacteria are true cells. Evidence: In our comparison of *E. coli* with a human cheek cell, the bacterium had a cell membrane, cytoplasm, DNA and ribosomes. Its DNA lies free in a region called the nucleoid instead of inside a membrane-bound nucleus, and it has no membrane-bound organelles. Reasoning: A structure counts as a cell because it has the four universal features, not because it has a nucleus. Bacteria such as *Lactobacillus* in fermenting uji have all four, so they are cells (prokaryotic cells), while human and sukuma wiki cells are eukaryotic because they have a membrane-bound nucleus. (c) The membrane controls what enters and leaves, the cytoplasm is where cell reactions happen, DNA carries the instructions, and ribosomes make proteins. Without these a cell could not carry out the basic processes of life such as protein synthesis, using energy and reproducing.

Part 2: Plant cell or animal cell?

Use what you saw in the onion and cheek cell practical and in the sukuma wiki leaf cell. (a) Describe three structures found in the sukuma wiki leaf cell but NOT in the human blood cells,

(a) Cell wall: a rigid layer made of cellulose outside the cell membrane that gives the cell support and a fixed shape. Chloroplasts: green organelles containing chlorophyll that capture light for photosynthesis, so the leaf can make its own food (glucose). Large central vacuole: a water-filled sac that creates turgor pressure to keep the cell firm and stores water; it pushes the other organelles to the edge of the cell. (b) The sukuma wiki cell has a rigid cellulose cell wall that holds it in a rectangular shape. The cheek cell has only a flexible cell membrane and no wall, so its shape is irregular and rounder. (c) The green colour comes from chlorophyll in chloroplasts. The onion layer we viewed did not show green, which suggests it lacks chloroplasts in that part of the plant;

and say what each one does. (b) Explain why the sukuma wiki cell looks rectangular but a cheek cell looks irregular and rounder. (c) The onion epidermal cells we viewed were not green, although onion is a plant. Suggest a reason, using what you learned about the organelle responsible for the green colour.	chloroplasts are found in cells that capture light, such as leaf cells, so the sukuma wiki leaf cell is green while the onion layer was not.
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Part 3: Structure and function in specialised cells

Look back at the blood smear from the Kisumu County Hospital. (a) The red blood cells have no nucleus and are biconcave discs. Explain how this structure is linked to the job of the red blood cell. (b) The white blood cells have large nuclei. Using what you learned about the nucleus, explain what the nucleus allows a cell to do that a mature red blood cell cannot do. (c) Choose ONE other specialised cell (guard cell, root hair cell, muscle cell, nerve cell or sperm cell). Describe its special structure and explain how it helps the cell do its job, with a Kenyan example from the sukuma wiki plant, a bean seedling or the human body.	(a) The red blood cell has no nucleus, which leaves more internal space for haemoglobin, so it can carry more oxygen around the body. Its job is oxygen transport, so losing the nucleus is a specialisation, not a defect. (b) The nucleus is the cell's control centre, like the Managing Director's office in our factory analogy. It holds the DNA and directs protein making and cell division. White blood cells keep their nuclei so they can make proteins and carry out their functions, while a mature red blood cell cannot make new proteins or divide. The trade-off is that the red blood cell gains oxygen-carrying capacity. (c) Example: the root hair cell is long and thin, which increases surface area for absorbing water, so a bean seedling in a shamba can take up water efficiently from the soil. (Other correct examples: guard cells are bean-shaped and open and close the stomata in a sukuma wiki leaf; muscle cells are elongated with many mitochondria to power contraction; nerve cells have long axons to carry signals quickly; sperm cells have a flagellum and a mitochondria-rich midpiece to swim to the egg.)
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Part 4: When structure changes, function changes

In sickle cell anaemia, some red blood cells have a crescent shape instead of the normal shape. Explain how the change in structure affects the function	Sickle cells have a crescent shape instead of the normal biconcave disc. The crescent cells disrupt the packing of haemoglobin and stack together and block capillaries, so less oxygen is delivered to the body tissues. This causes serious health problems for the affected person, including tiredness and pain. This shows that structure and function are closely linked: a cell is shaped to do a particular job, and even a small structural change caused by a single gene mutation can disrupt that job in measurable, life-affecting ways. This is an important health issue in Kenyan communities where the sickle cell trait is present.
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of the cell and what this means for a person's body. Then explain, using this example, why structure and function are linked in cells.	
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Part 5: Answer the driving question

Write a full explanation to answer the driving question: Why do the cells in my body, and in the sukuma wiki on my plate, look so different from each other, and how does each cell's structure allow it to do its specific job? Your answer must include: (1) the structures all cells share, (2) the main differences between plant and animal cells, (3) the idea of specialisation, explaining what is added, removed or modified, and (4) one Kenyan example that shows structure matching function.	All cells, whether red blood cells, white blood cells, sukuma wiki leaf cells or bacteria, share a cell membrane, cytoplasm, DNA and ribosomes, because these are needed for the basic processes of life. This is why they are all called cells. Plant cells such as the sukuma wiki leaf cell also have a cellulose cell wall for support and rectangular shape, chloroplasts for photosynthesis and a large central vacuole for turgor pressure and water storage. Animal cells like our blood cells lack these, though animal cells may have centrioles for cell division. Cells look different because of specialisation: a specialised cell is not a faulty generic cell but one that is modified for a job. Structures that are not needed are removed or reduced, and structures that help the job are added or amplified. For example, the red blood cell in the Kisumu blood smear loses its nucleus to make room for haemoglobin so it carries more oxygen, while the white blood cell keeps its nucleus. The sukuma wiki leaf cell has chloroplasts because it must harvest light from the sun, which an animal cell never needs to do. Likewise, root hair cells of a bean seedling are long to absorb water. So the cells look different because each one is built, in a specific organism and environment, to do its own specific job, and the shared structures show they are all cells.
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Shared cell features and cell types	Names all four universal features (membrane, cytoplasm, DNA, ribosomes), correctly distinguishes prokaryotes from eukaryotes and uses a full CER with evidence and reasoning.	Names most universal features and states that bacteria are cells, but evidence or reasoning is incomplete.	Names few shared features or confuses what makes something a cell.
Plant versus animal cell structures	Correctly names cell wall, chloroplasts and central vacuole, explains the function of each and links them to the shape and colour of the sukuma wiki cell.	Names the plant structures with some functions, with limited links to the observed cells.	Lists structures without functions or mixes up plant and animal features.
Structure-function reasoning in specialised cells	Explains clearly how the structure of the red blood cell, white blood cell and another specialised cell suits its job, including the trade-off of losing the	Links structure to function correctly for most examples but with limited detail.	States structures or functions separately without linking them.

	nucleus.		
Use of evidence and Kenyan context	Draws consistently on the blood smear, sukuma wiki, practicals and sickle cell example, accurately and in the same way as taught.	Uses some evidence and examples from the lessons.	Uses little or no evidence from the lessons, or inaccurate examples.
Communication of the final explanation	Answer to the driving question is organised, complete (all four required parts), and uses scientific vocabulary accurately.	Answer covers most required parts with mostly correct vocabulary.	Answer is incomplete or unclear, with frequent vocabulary errors.

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